Avery Briggs | abriggs914@gmail.com | github.com/abriggs914

Professional Summary

Microsoft Access database developer and systems analyst with 5+ years of experience designing and maintaining database applications supporting manufacturing and operational workflows. Experienced in full database lifecycle development including relational schema design, form and report development, query optimization, VBA automation, and SQL Server integration. Known for building tools that improve data visibility, automate manual processes, and support operational decision-making. Comfortable working remotely, managing independent projects, and collaborating with stakeholders to translate business requirements into practical database solutions.

Experience

Software Systems Developer | Database Developer

BWS Manufacturing Ltd. | Centreville, N.B. (Jan 2021 – Feb 2026)

Developed internal software systems and software development tools supporting operational workflows across Sales, Procurement, Production, and Scheduling within a small technical team in a manufacturing environment.

Key Contributions

- Researched and validated operational datasets to ensure accuracy across inventory, procurement, and manufacturing systems.
- Developed analytical queries and reporting tools enabling cross-department analysis of operational data including inventory, production, purchasing, and sales metrics.
- Refactored legacy Microsoft Access applications by consolidating duplicated VBA logic into reusable shared modules, eliminating 3–5 smaller databases and improving maintainability of the primary database system.
- Developed VBA automation for validation, reporting, and data entry tools within Access applications.
- Conducted structured data analysis using SQL, Python, and reporting tools to identify trends and anomalies.
- Built tools that improved data visibility and reduced reconciliation errors across multiple departments by verifying accuracy of data through cross-referencing sources and historical records.
- Designed a warehouse data visualization system mapping inventory locations onto spatial layouts to improve operational visibility and planning.
- Developed tools using OCR techniques to extract structured data from legacy PDF documents, enabling automated reporting and analysis of previously inaccessible operational records.
- Built an administrative time-tracking interface integrating live ShopClock data, allowing supervisors to review, adjust, and approve employee shift records before payroll submission, improving visibility and coordination across multiple departments.
- Provided technical troubleshooting and support across hardware systems, network devices, ERP integrations, and vendor-managed services.

Notable Projects:

- IT Request and Inventory Ticketing System. Built from scratch using SQL and Access.
- Developed an Access macro and VBA workflow to automate batch report generation, triggered from a Python/Streamlit dashboard. Implemented timestamp validation and fault-tolerant logic to ensure reliable generation of up to 50 reports per execution.
- Yellow Tag system to automatically flag, record, and report missed parts when picked for production. Offers a variance reconciliation opportunity earlier in the production pipeline, highlighting common trends. Front-end built using an automatically refreshing form presenting live Syspro Data.
- PDF parsing of legacy reports using Python and modules: Pdfplumber, Streamlit, and Pytesseract.
- 3D warehouse visualization using Plotly, demonstrating bottlenecks and improving transparency.

Technologies used:

- Microsoft Access, VBA, SQL Server, Syspro8 ERP, ShopClock, Python, Streamlit, Pandas, Plotly, Pytesseract, Reportlab, CustomTkinter, NeWare Security

Other Job Experience:

- Basketball Referee – CBOC – Fredericton, NB – Oct 2013 – Mar 2020
- Kitchen Staff – FMI – Fredericton, NB, -- May 2015 – Sept 2017
- Farm Hand – Hunter Brother's Farm Market – Florenceville, NB, -- Jun 2011 – Aug 2013

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Education

Bachelor of Computer Science

UNB – Fredericton N.B. (May 2020)

- Specialization: Software Systems.
- Recipient of Governor Thomas Carleton Entry Scholarship.

Skills

- **Languages**: English (native), French (Intermediate professional proficiency)
- **Programming**: Python, VBA, SQL, Java
- **Databases**: Microsoft Access, Microsoft SQL Server
- **Frameworks / Libraries**: Streamlit, Pandas, Plotly, CustomTkinter, Pdfplumber, Pytesseract
- **Learning**: TypeScript, C#, .NET, other modern web frameworks
- **APIs**: NHL Stats API, IMDB API, Fitbit API
- **Tools:** Git, Syspro ERP, ShopClock
- **Other**: Strong cross-team communication and collaboration. Self-directed problem solving and rapid learning.

Awards & Activities

- Finalist (Top 12) - Kaspersky Academy Partner's Summit (KAPS) Lisbon, Portugal (December 2019).
- Certified intermediate plus in the French oral proficiency exam (June 2013).
- UNB dormitory Orientation Committee (September 2014) & Social Representative (September 2014 – April 2015).
- UNB Relay for Life steering committee logistics team (September 2013 – February 2014).

Hobbies

3D printing and hardware projects, sports analytics software, recreational athletics (cycling, kayaking, basketball officiating)

Projects

- NHL data analysis dashboard displaying real-time game data and standings using the NHL Stats API.
- STL file generation for automating the creation of strings of text using custom fonts and other parameters. Uses a Solid2 to generate files, and Streamlit interface to model and tweak them.
- Minesweeper Android mobile application, built using Java and Android Studio.
- Image editing Android application taking advantage of the Gestures library to offer a unique interface for tuning photos.
- Traffic simulation engine using Pygame. Mimics some elements from Cities Skylines traffic patterns.
- COVID-19 interaction simulator built with JavaFX modeling contagion spread in configurable populations.

References available upon request